

CANDIDATE NAME \_\_\_\_\_

INDEX NUMBER \_\_\_\_\_

CG \_\_\_\_\_



SERANGOON JUNIOR COLLEGE  
JC2 Preliminary Examination 2016

H2 BIOLOGY  
9648

|                |
|----------------|
| <b>Paper 2</b> |
|----------------|

2 Hours

# MARK SCHEME

Date / Day: /

## INSTRUCTIONS TO CANDIDATES

Write your name, CG and index number in the spaces at the top of this page and on all separate writing papers used.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams, graphs or rough working.

### Section A

Answer **all** questions.

Write your answers in spaces provided on the question paper.

### Section B

Answer **only one** question out of two.

Write your answers on the separate answer paper provided.

## INFORMATION FOR CANDIDATES

The intended number of marks is given in brackets [ ] at the end of each question or part question.

### FOR EXAMINER'S USE

| Section A |      |
|-----------|------|
| 1         | /11  |
| 2         | /    |
| 3         | /    |
| 4         | /13  |
| 5         | /    |
| 6         | /    |
| 7         | /10  |
| Total     | /80  |
| Section B |      |
| 9 OR 10   | /20  |
| TOTAL     | /100 |

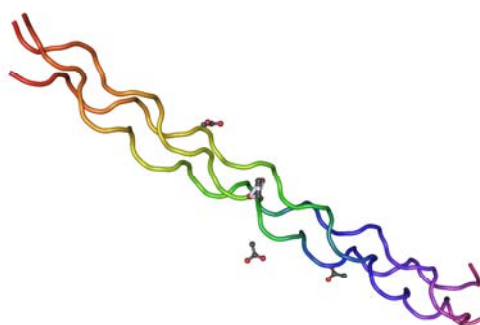
This question paper consists of 23 printed pages and 1 blank pages

## SECTION A

Answer **all** questions.

### Question 1

**Figure 1.1** below depicts the molecular structure of a basic unit of collagen.



**Figure 1.1**

(a) State the name given to such a basic unit of collagen. [1]

- Tropocollagen.

(b) Describe how the monomers of this basic unit are joined together to achieve the final molecular configuration as shown in Figure 1.1. [3]

- Monomers are amino acids
- Join together via condensation reaction to form peptide bonds with a loss of water molecules forming a polypeptide
- Three such polypeptides join together via intermolecular hydrogen bonding between NH and CO groups to form tropocollagen

Collagen is normally found in animal connective tissue where its role is as a structural molecule. Figure 1.2 shows an electron micrograph of collagen.



**Figure 1.2**

(c) Explain the banded appearance of collagen shown in Figure 1.2. [2]

- Due to **staggered arrangement/ longitudinal displacement** of tropocollagen subunits with respect to each other.
- Held by **covalent cross links** between adjacent tropocollagen subunits.

(d) Explain why collagen is able play the role of a structural molecule. [2]

- Large molecular size so insoluble.
- large **number of hydrophobic, non-polar amino acids** that face the outside of the triple helix so insoluble

(Max 1 for solubility)

- Several tropocollagen molecules are further **covalent cross-linked** with neighbouring tropocollagen molecules running parallel to them to form a collagen fibrils/fibres giving great tensile strength.

Another common structural molecule found in nature is cellulose. Cellulose is the main structural molecule in plants.

(e) Compare the structures of cellulose and collagen. [3]

Similarities:

- Both have intermolecular hydrogen bonding
- Both associate to form fibrous structures.

Differences:

- $\beta$ -glucose monomer in cellulose vs amino acid monomer in collagen
- $\beta$  1,4 Glycosidic bond in cellulose vs peptide bond in collagen.
- Cellulose is a straight chain structure while collagen is helical

[Total: 11 marks]

## Question 2

The following Figure 2.1 shows an electron micrograph of several cells.

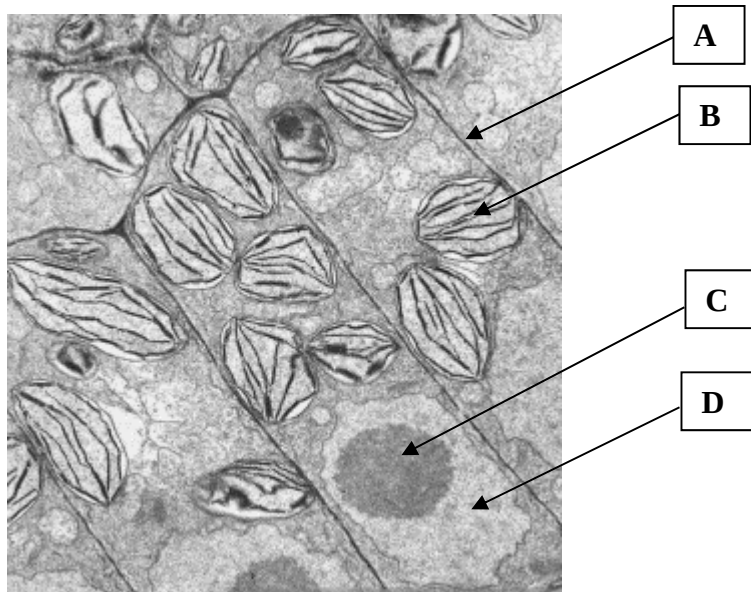


Figure 2.1

(a) Label the organelles A – D. [4]

A – Cell Wall

B - Chloroplast

C - Nucleolus

D – Nucleus

(b) Discuss the role of organelle C. [3]

- Contains rRNA genes
- Involved in the **synthesis of ribosomal RNA (rRNA) via transcription**,
- which is a constituent of **ribosomes**.
- **Also involved in the assembly of rRNA and ribosomal proteins into large and small subunits**

Max 3

(c) In the nucleoplasm of such cells, genetic material can be found. State the nature of this genetic material and briefly describe how this genetic material is organized. [3]

- Made up of DNA
- Organised into chromosomes
- Condensed around histone octamer and non-histone proteins

Such genetic material can take part in processes such as what is shown in **Figure 2.2**.



**Figure 2.2**

(d) With appropriate reasons, precisely identify the process shown in **Figure 2.2**. [3]

- Mitosis Metaphase
- Chromosome lined up at the equator of the cell
- No nuclear envelope visible
- Only one row and not two so not meiosis

Max 3

(e) Suggest the significance of the **next stage** of the above process. [2]

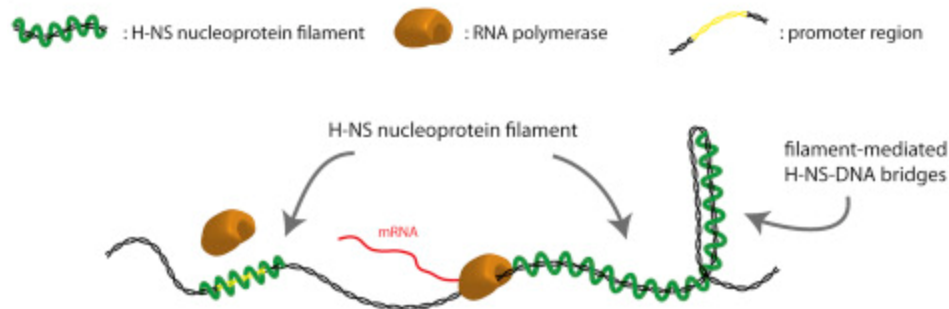
- Next stage is anaphase (no credit for stating this)
- Separates sister chromatids into individual chromosomes
- Allows each replicated chromatid/chromosome to move to the pole of the cell
- Ensures that each daughter cell will have the original chromosome number restored.
- AVP

Max 2

[Total: 15 marks]

### Question 3

A set of abundant nucleoid-associated proteins (NAPs) play key functions in organizing the bacterial chromosome and regulating gene transcription globally. Histone-like nucleoid structuring protein (H-NS) is representative of a family of NAPs that are widespread across bacterial species. They have drawn extensive attention due to their crucial function in gene silencing in bacterial pathogens. Figure 3.1 illustrates how H-NS is able to silence genes of bacterial pathogens. (Information obtained from *Biophysical Journal* Oct 2015, 109(7))



**Figure 3.1**

(a) State the name of one bacterial pathogen. [1]

- Bacteriophage

(b) Explain the role of RNA polymerase in a bacterial cell. [3]

- Involved in transcription, leading to the production of messenger RNA.
- Messenger RNA acts as a template for translation into proteins at ribosomes.
- Binds to promoter of gene, bringing ribonucleotides together, joining them via phosphodiester bonds

(c) Using the information from Figure 3.1, suggest how H-NS is able to silence genes. [2]

- Inhibits binding of RNA polymerase to the promoter region
- Blocks the movement of RNA polymerase along the gene

The following figure shows regulation of transcription in the *trp* operon of a bacterial cell.

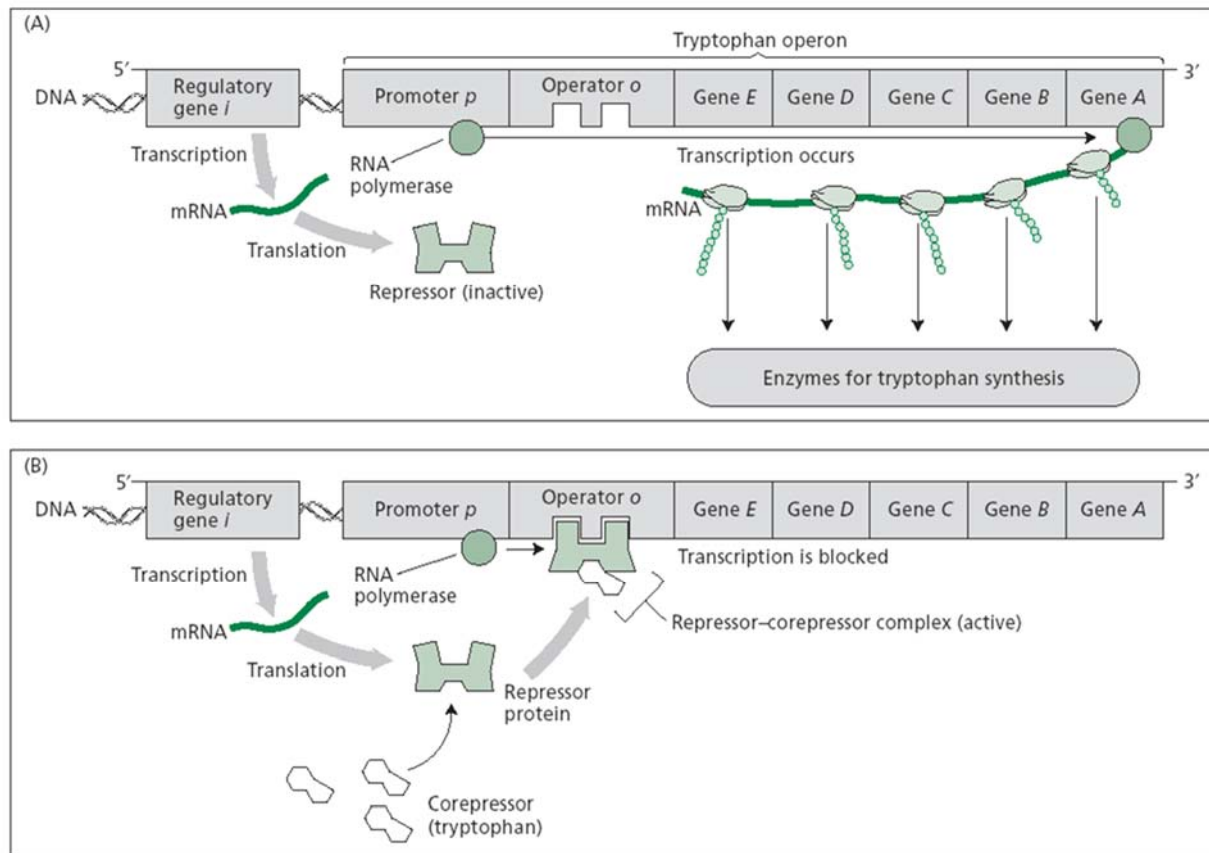


Figure 3.2

(d) Using the information in figure 3.2, describe the process of negative regulation in bacterial operons. [4]

- Involves the use of a repressor molecule
- In the *trp* operon example, repressor is made in the inactive form, operon default on/transcription occurs.
- In the presence of co-repressor tryptophan, repressor is activated = repressor-corepressor complex.
- repressor-corepressor complex binds to operator, block RNA polymerase and operon is switched off/transcription does not occur.

(e) Suggest why tryptophan is the ideal co-repressor for this operon. [2]

- Enzymes produced by operon result in the synthesis of tryptophan.
- Tryptophan thus inhibits its own production when it is in excess.
- Acts as an end product inhibitor

Max 2

- (f) The lac operon is an example of an inducible operon system. Contrast negative regulation of the lac operon with that of the trp operon. [2]

| Lac operon                            | Trp operon                              |
|---------------------------------------|-----------------------------------------|
| Repressor produced in the active form | Repressor produced in the inactive form |
| Requires allolactose as an inducer    | Requires tryptophan as a co-repressor   |

#### Question 4

The diagram shows the chromosome from a plant cell in which  $2n = 4$ . The dominant allele **A** of a gene results in red flower and pink spines on the fruit. The recessive allele, **a**, gives yellow flower and green spines. The dominant allele **B** intensifies the colour of the pink spines to red. The plant is heterozygous as shown below:

- (a) The plant was selfed. Give the phenotypic ratio of the offspring. Explain your answer. [3]

- 9 red flower and red spines on the fruit : 3 red flower with pink spines on the fruit : 4 yellow flower with green spines on the fruit;
- Due to epistasis;
- Offspring with genotype aa and B/b alleles will have yellow flower and green spines on the fruit as pink is not expressed so the colour cannot be intensify even in the presence of B allele/ recessive alleles a **mask** the dominant allele B;

The plant is also heterozygous for another characteristic, the height of the plant. Allele T = tall is dominant to allele t = dwarf. A cross was made between the tall, red-flowered plant with a dwarf, yellow-flowered plant and produced a large number of offspring. **Table 4.1** shows the results.

**Table 4.1**

| Phenotype                    | Number of offspring |
|------------------------------|---------------------|
| Tall, red-flowered plant     | 35                  |
| Tall, yellow-flowered plant  | 14                  |
| Dwarf, red-flowered plant    | 16                  |
| Dwarf, yellow-flowered plant | 33                  |



- (b) The ratio of phenotypes expected in a cross such as this is 1:1:1:1. Chi-squared test was performed on these data giving a calculated value of  $X^2$  of 14.64.

**Table 4.2:** Distribution of  $X^2$

| Degrees of freedom | Probability, p |      |       |       |       |
|--------------------|----------------|------|-------|-------|-------|
|                    | 0.10           | 0.05 | 0.02  | 0.01  | 0.001 |
| 1                  | 2.71           | 3.84 | 5.41  | 6.64  | 10.83 |
| 2                  | 4.61           | 5.99 | 7.82  | 9.21  | 13.82 |
| 3                  | 6.25           | 7.82 | 9.84  | 11.35 | 16.27 |
| 4                  | 7.78           | 9.49 | 11.67 | 13.28 | 18.47 |

- (i) Use the calculated value of  $X^2$  and the table of probability provided in **Table 4.2** to find the probability of the results of the cross is due to chance. [1]

1. Probability:  $0.001 < p < 0.01$

- (ii) State what conclusions may be drawn from the probability found in (b)(i). [2]

1. There is significant difference between expected results and actual results/ Probability of the actual results deviating from the expected results due to chance is low since it is less than 0.05
2. Inheritance pattern does not conform to the 1:1:1:1 ratio

- (iii) Explain the difference between the expected and actual results of the test cross. [3]

1. Difference is due to the two genes being linked together
2. Tendency for these two genotypes (AT and at) of the gametes to be produced → more tall, red-flowered plant and dwarf, yellow-flowered plants are produced
3. Due to crossing over during Prophase I of meiosis, new recombinants where A is linked to t and a is linked to T are produced, resulting in tall, yellow-flowered plants and dwarf, red-flowered plants.

- (c) Briefly explain how another factor, besides crossing over, can bring about genetic variation in the offspring produced by a sexually reproducing organism (**Exclude mutation from your answer**). [1]

1. Independent assortment of homologous chromosomes during meiosis 1 results in different combination of alleles into the gametes /  $2^n$  possible combinations where n is the number of homologous pairs. (Rej: independent assortment without any sufficient elaboration)

OR

2. random fertilization/ fusion of gametes results in different genotypes/ combination of alleles in the offspring/ results from each gamete having different sets of chromosomes

In another completely different cross involving humans, the following results were obtained.

| Phenotypes      | Genotypes | Color of skin |
|-----------------|-----------|---------------|
| Extremely dark  | PPQQRR    | 6             |
| Very dark       | PpQQRR    | 5             |
| Dark            | PpQqRR    | 4             |
| Intermediate    | PpQqRr    | 3             |
| Light           | ppQqRr    | 2             |
| Very light      | ppqqRr    | 1             |
| Extremely light | ppqqrr    | 0             |

(d) What is the term used to describe the range of phenotypes in the above example? [1]

1. Continuous Variation (Rej: Polygenic inheritance)

(e) Explain the genetic basis for the range of phenotypes seen. [2]

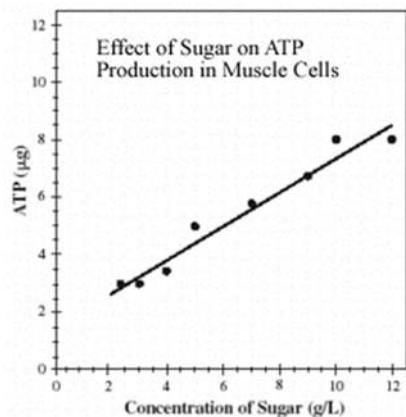
1. Human skin colour is controlled by **3 pairs/many** of independent genes. (rej: multiple alleles)

2. Cumulative/additive effect on the trait without complete dominance.

[Total: 13 marks]

## Question 5

The following figure shows the effect of sugar levels on muscle cells.



(a) Describe and explain the effect of sugar on ATP production in muscle cells. [3]

1. As concentration of sugar increases from 2 to 12g/L ATP production increases from 2.5 to 8.5ug.
2. Sugar (Glucose) used in mitochondria of muscle cells in aerobic respiration/ description of stages of respiration **apart from only glycolysis**
3. Aerobic respiration produces ATP

(b) Suggest and explain what will happen to ATP production if the concentration of sugar continued to increase. [2]

1. It will plateau off
2. Possible limiting factors with example = rate of sugar uptake into muscle cell/ respiratory enzyme conc/ oxygen concentration AVP (Rej: limited number of cells or mitochondria)

The coenzyme cytochrome c oxidoreductase, is the third complex in the electron transport chain, playing a critical role in biochemical generation of ATP (oxidative phosphorylation).

(c) Explain how the coenzyme cytochrome c oxidoreductase may aid in the production of ATP in the electron transport chain. [3]

1. Serves as an electron carrier
2. Energy released when passing electrons used to pump/ actively transport H<sup>+</sup> into intermembrane space/ create proton gradient
3. H<sup>+</sup> then moves down a conc gradient through the stalked particle/ ATP synthase complex to make ATP from ADP and Pi.
4. AVP

(d) Actimycin A is known to inhibit the coenzyme cytochrome c oxidoreductase. Predict the effect actimycin A would have on the aerobic respiratory process. [2]

1. Electrons cannot move down the chain, ATP not made
2. NAD<sup>+</sup> and FAD not regenerated
3. hence Krebs cycle and link reaction shut down, aerobic respiration shut down
4. AVP

Rej: inhibit ATP production unless accompanied by valid explanation.

Any 2

[Total: 10 marks]

The following Figure 6.1 shows the synaptic knob of a synapse found in the sympathetic ganglion of frogs. Such synapses are unique as they contain receptors to the neurotransmitter noradrenaline (NE) on the pre-synaptic membrane itself. These are known as autoreceptors.



- 13

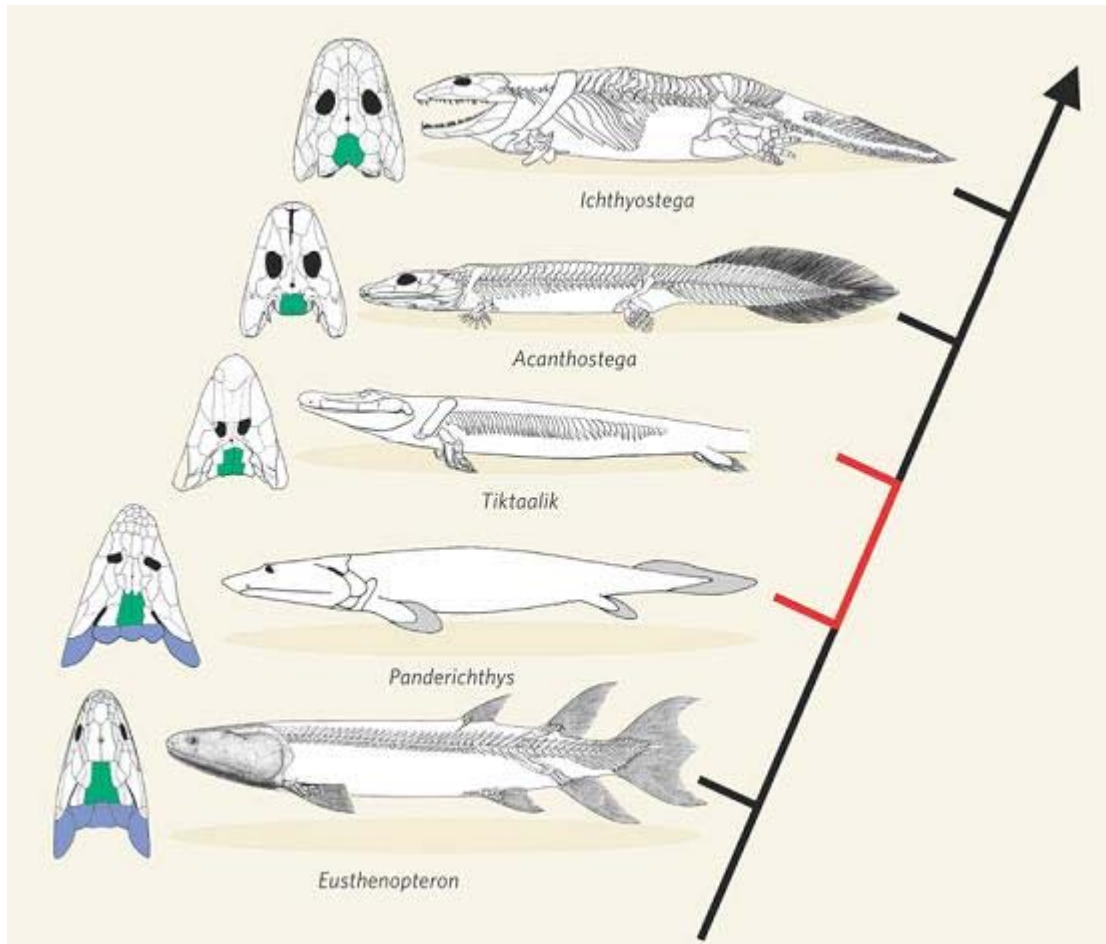
(d) Explain one function of synapses. [2]

1. Ensure **one-way transmission of nerve impulses**. As only the presynaptic membrane can release neurotransmitter and only postsynaptic membrane has receptors that trigger an action potential.
2. The post-synaptic neurone and synapses act as **a junction for the integration of stimuli** from **various sources** to **produce a co-ordinated response**. (Allow for alternative point on inhibition)
3. **Filter out** low-level stimuli that are of no significance, i.e. remove “background noise” from the body’s nervous system as all neurotransmitter will have been released but stimuli continues = adaptation

[Total: 7 marks]

## Question 7

Figure 7.1 shows a series of fossils. This series depicts how land-based amphibians could have evolved from fishes. *Tiktaalik* hails from the Late Devonian period, about 360 million years ago, and is both chronologically and morphologically intermediate between two other major fossils in this series, the more fish-like *Panderichthys* and the more tetrapod-like *Acanthostega*.



- (a) State the name given to fossils such as *Tiktaalik* that have characteristics from two seemingly diverse groups of organisms. [1]
- Transitional forms
- (b) Explain how such fossil records can actually support Darwin's theory of evolution. [4]
- Similarities amongst group of organisms
  - Suggests a common ancestor
  - Differences amongst group of organisms
  - Suggests modification due to natural selection
  - Evidence for descent with modification.

In most respects *Tiktaalik*'s body is fish-like: it has fins and gill arches, just like a fish. However, its skull and especially its limbs mark it as a tetrapod ancestor. Species such as *Panderichthys* had true fins, similar to those of modern ray-finned fishes, consisting of an array of long, thin, spindly bones unsuitable for bearing weight. On the other side of the gap

is *Acanthostega*, with true limbs – each containing a radius and an ulna, just like our arms, and outfitted with eight true toes.

(c) State the name given to similar structures such as the limbs of *Acanthostega* and *Tiktaalik*.

- Homology / Homologous structures

(d) It was believed that the environment *Tiktaalik* evolved in was filled with swampy, silty lagoons. These dirty, unclear water masses also tended to have algae covering its surface. Using this information, describe how the amphibian-like *Acanthostega* could have evolved from the species *Tiktaalik*. [4]

- Variation existed within the *Tiktaalik* population; some individuals had the beginnings of true limbs.
- Formed two sub-populations, one on land and one in the swamps (those with fins and gill arches)
- These individuals within the swamps were selected against as the silty water and algae made survival more difficult, eg difficulty breathing. OWTTE/ Those on land selected for as easier to breathe/more food options OWTTE
- Alleles for true limbs passed down to next generation.
- No interbreeding/genetic isolation/reproductive isolation existed between the two sub-populations

Stimulus information adapted from:

<http://www.patheos.com/blogs/daylightatheism/2006/04/hello-beautiful/> &  
<https://sciencenotes.wordpress.com/tag/>



## Section B

### Question 8

**(a)** Explain the eukaryotic processing of pre-mRNA. [6]

1. Capping changes 5' end with the addition of 7-methylguanosine group;
2. Protects mRNA from degradation by 5' exonucleases / confers stability / assists in ribosomal binding;
3. Splicing removes introns and ligates exons to produce a mature mRNA;
4. Alternative splicing may occur, resulting in different combinations mature mRNA from a gene;
5. Addition of poly-A tail to 3' end of mRNA;
6. Allows slower degradation by 3' exonucleases / longer poly(A) tail implies longer half-life of mRNA / facilitates transport of mRNA from nucleus to cytoplasm;

**(b)** Compare and contrast between prokaryotic and eukaryotic control of gene expression at the translational and post-translation level. [9]

| Feature                   | Eukaryotic                                                                                                  | Prokaryotic                                                                             |
|---------------------------|-------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| Initiation of translation | Initiated by recognition of AUG sites by small ribosomal subunit;                                           |                                                                                         |
|                           | First amino acid methionine                                                                                 | First amino acid N-formyl-methionine;                                                   |
| Translational repressors  | Translational repressors bind at 5' UTR to prevent translation initiation;                                  |                                                                                         |
| Stability of mRNA         | Eukaryotic mRNA more stable / present in cytoplasm for longer period of time, due to 5' cap and poly A tail | Prokaryotic mRNA has no poly A tail or 5' cap, so less stable / have shorter half-life; |
| Biochemical Modification  | Addition of biochemical groups to proteins → activate protein;                                              | Biochemical modification not as significant due to lack organelles such as ER & GA;     |
| Protein degradation       | Unwanted / misfolded proteins can be degraded by proteasomes / lysosomal degradation                        | Lack lysosomes / proteasomes but possess other mechanism of degrading proteins;         |
| Feedback inhibition       | Feedback mechanism present to control gene expression;                                                      |                                                                                         |

- (c) Describe the significance of gene amplification. [5]
1. Definition – selective increase in number of copies of a particular gene without a proportional increase in other genes;
  2. Meet the needs of cells resulting in higher level of mRNA and polypeptide synthesis at different development stages of cells;
  3. Specific example
  4. May cause diseases like cancer due to over-expression of proteins leading to development of malignant tumours ;
  5. May confer drug resistance, ref to example of methotrexate;
  6. May confer selective advantage which allow organisms to survive in a particular environment;
  7. May contribute to evolution of genome, ref to homologous genes;

### Question 9

(a) With reference to the three different stages of cell signaling, describe in detail the sequence of events when a glucagon molecule reaches the liver cell. [10]

#### **Signal reception**

***The hormone, glucagon acts the “first messenger” or a ligand, binding specifically to a G-protein-linked receptor located on the outside surface of the plasma membrane of the target cell.***

***Forming a hormone receptor complex***

***activates the G protein → displacement by GTP of GDP***

***The activated G protein moves to and then activates the enzyme adenylyl cyclase, Adenylyl cyclase catalyses the formation of cAMP from ATP.***

#### **Signal transduction**

***cAMP acts as a second messenger***

***moves within the cell to activate enzymes***

***such as protein kinases.***

***Protein kinases then activate other enzymes by phosphorylating them.***

***In a cascade reaction***

***Signal is amplified***

#### **Cellular response**

***regulate cellular activities in the cytoplasm by increasing the blood glucose level to 90mg glucose/100ml blood***

***regulate transcription in the nucleus***

***by synthesis of enzymes or other proteins from genes***

***eg. activating enzymes required for increasing blood glucose level***

***such as glycogenolysis – breakdown of glycogen to give glucose,***

***gluconeogenesis – synthesis of glucose from sources other than carbohydrates, etc***

(b) Discuss roles played by proteins in maintaining the potential differences across membranes and in the transmission of nerve impulses along an axon [7]

1.  $\text{Na}^+/\text{K}^+$  pump on axon membrane;
2. to maintain membrane potential (potential difference across membranes);
3. via active transport requiring ATP;
4. for every 3  $\text{Na}^+$  out, 2  $\text{K}^+$  in;
5. to maintain an electrochemical gradient of  $\text{Na}^+$  and  $\text{K}^+$  across axon membrane
6. non-gated  $\text{K}^+$  and  $\text{Na}^+$  ion leakage channels;
7. based on facilitated diffusion controlling movement of  $\text{Na}^+$  &  $\text{K}^+$  in and out of axoplasm down electrochemical gradient;
8. there are 20x more  $\text{K}^+$  leakage channels in the axon membrane than  $\text{Na}^+$  channels
9. Both  $\text{Na}^+/\text{K}^+$  pump and leakage channels are responsible for the resting potential of the nerve cell
10. voltage gated  $\text{Na}^+$  and  $\text{K}^+$  ion channels;
11. nec for depolarisation and repolarisation of axon membrane
12. arrival of stimulus or initial depolarisation → opening of  $\text{Na}^+$  channels,
13. entry of  $\text{Na}^+$  into axoplasm causing depolarization of membrane
14. A.P. generated when threshold potential exceeded;
15. A.P. transmitted/propagated to adjacent region of axon by closure of  $\text{Na}^+$  gates so that  $\text{Na}^+$  must move to region ahead and not out of nerve cell;
16. efflux of  $\text{K}^+$  out of axoplasm as  $\text{Na}^+$  channels close &  $\text{K}^+$  channels open;
17. repolarisation occurs;

(c) Explain how loss of myelination, which happens in the demyelinating disease multiple sclerosis, disrupts signal transmission in the nervous system. [3]

***Myelin sheath made of lipids is important in electrical insulation***

***limiting the APs to the node of Ranvier allowing the A.P to jump from node to node increasing the speed of transmission***

***Loss of myelination leads to more AP generated → Decreasing/slow speed of transmission***

Accept the reverse explanation.